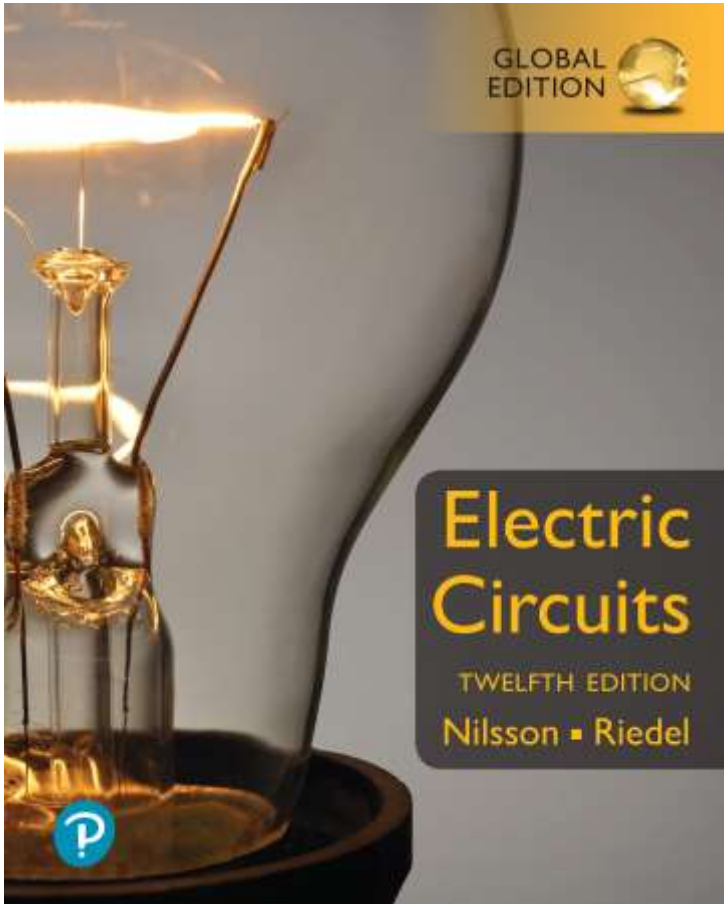
Electric Circuits

Twelfth Edition, Global Edition



Chapter 7 – Part 5

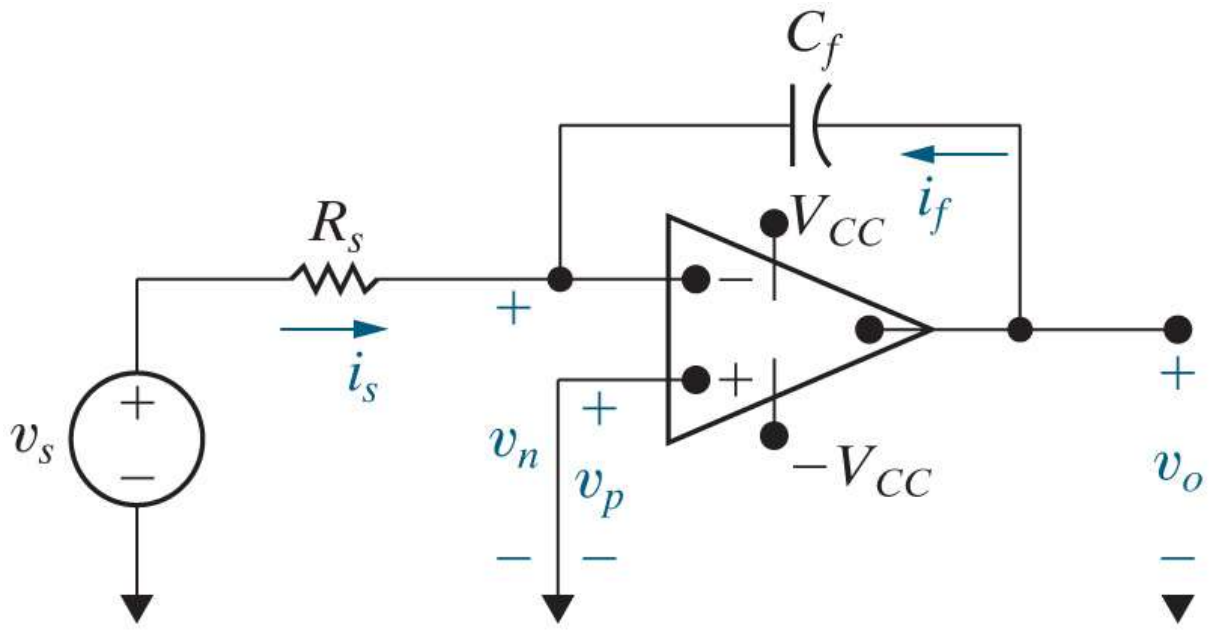
Response of First-Order *RL* and *RC* Circuits

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Chapter 7 – Integrating Amplifier (1 of 6)

The integrating amplifier, shown here, is another example of a first-order circuit, because its output voltage is described by a first-order differential equation.

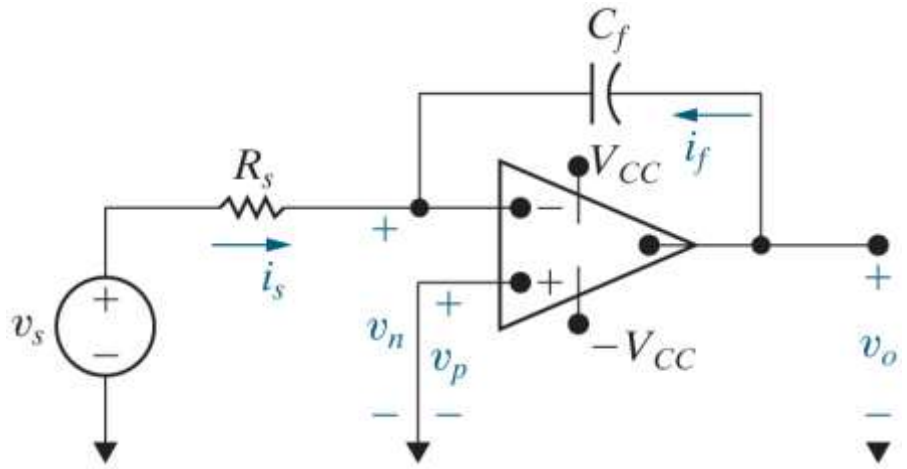


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Chapter 7 – Integrating Amplifier (2 of 6)

To analyze this circuit, we write a KCL equation at the op amp's inverting input node. Assuming the op amp is ideal, the voltage at this node with respect to the reference node is



A. V_s

B. V_o

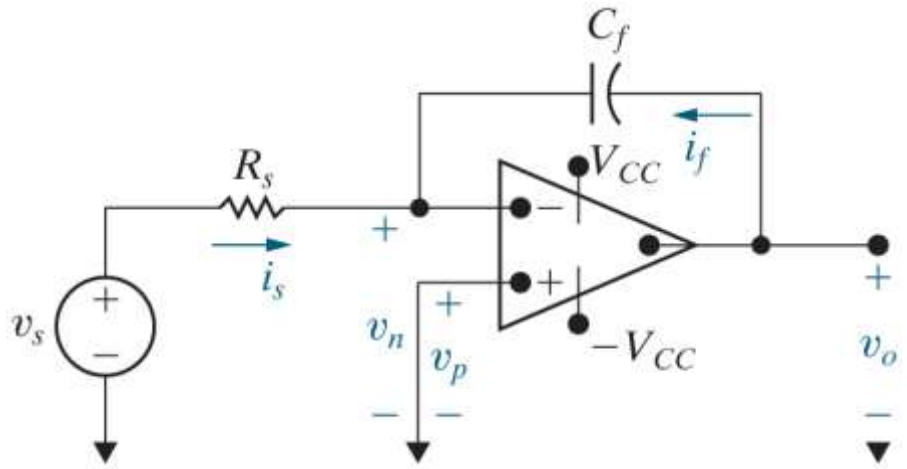
C. 0

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Chapter 7 – Integrating Amplifier (2 of 6)

To analyze this circuit, we write a KCL equation at the op amp's inverting input node. Assuming the op amp is ideal, the voltage at this node with respect to the reference node is



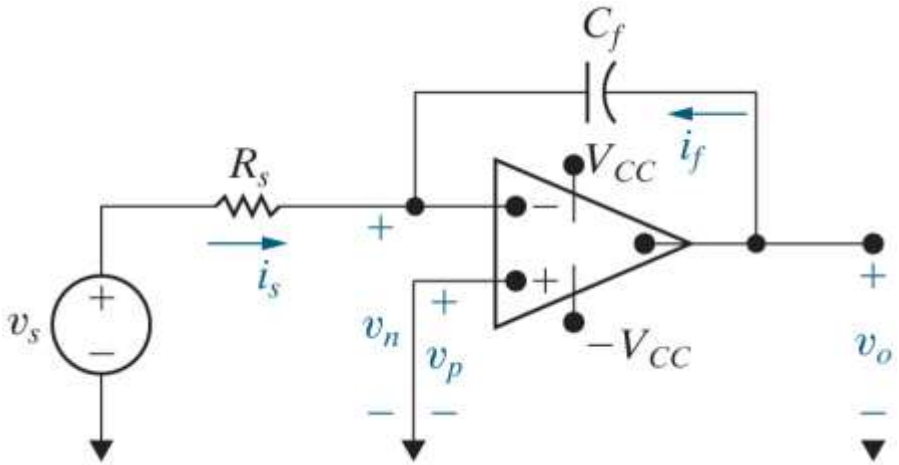
- ✗ A. V_s
- ✗ B. V_o
- ✓ C. 0

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Chapter 7 – Integrating Amplifier (3 of 6)

To analyze this circuit, we write a KCL equation at the op amp's inverting input node. Assuming the op amp is ideal, the current leaving this node and entering the op amp is



A. 0

B. i_s

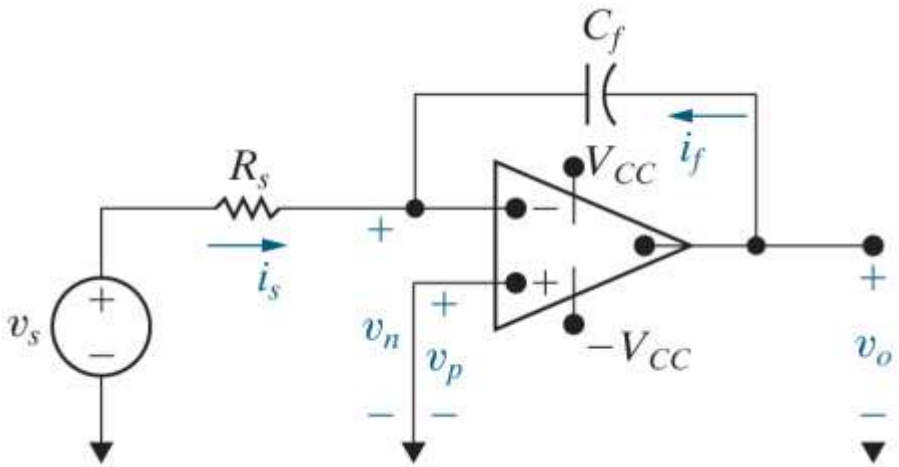
C. i_f

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Chapter 7 – Integrating Amplifier (3 of 6)

To analyze this circuit, we write a KCL equation at the op amp's inverting input node. Assuming the op amp is ideal, the current leaving this node and entering the op amp is

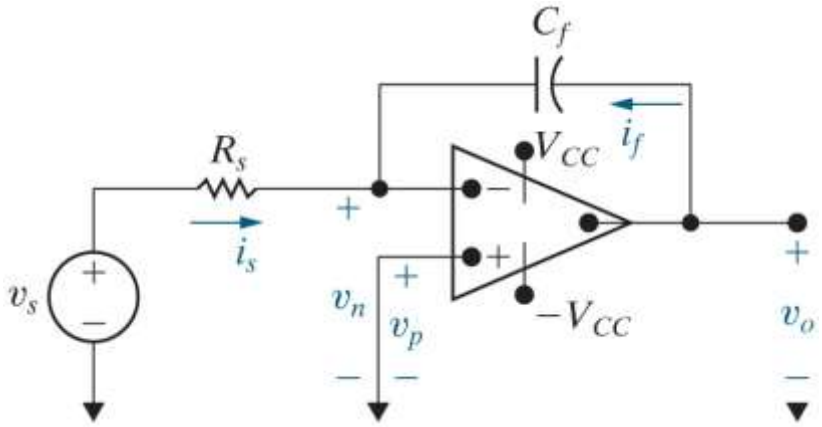


- ✓ A. 0
- ✗ B. i_s
- ✗ C. i_f

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Chapter 7 – Integrating Amplifier (4 of 6)



KCL at v_n: $$i_f + i_s = 0 \quad \Rightarrow \quad C_f \frac{dv_o}{dt} + \frac{v_s}{R} = 0$$

Solving for v_o: $$v_o(t) = -\frac{1}{R_s C_f} \int_{t_0}^{t} v_s \, dt + v_o(t_0)$$

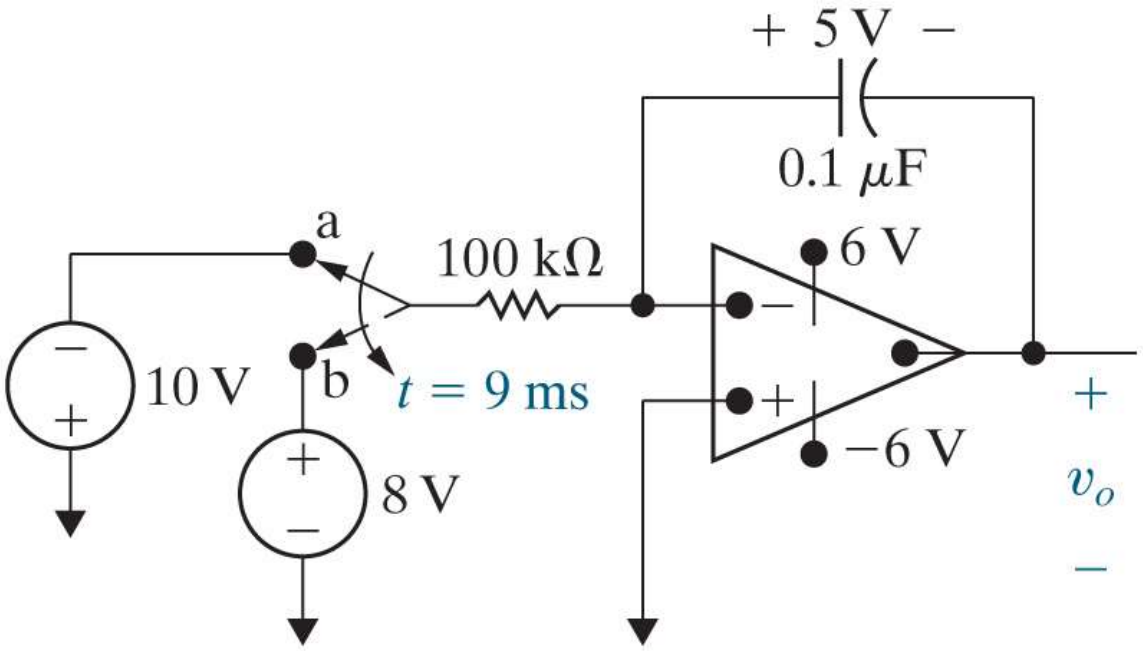
So the output of the integrating amplifier is proportional to the integral of the input, limited by the power supply voltages.

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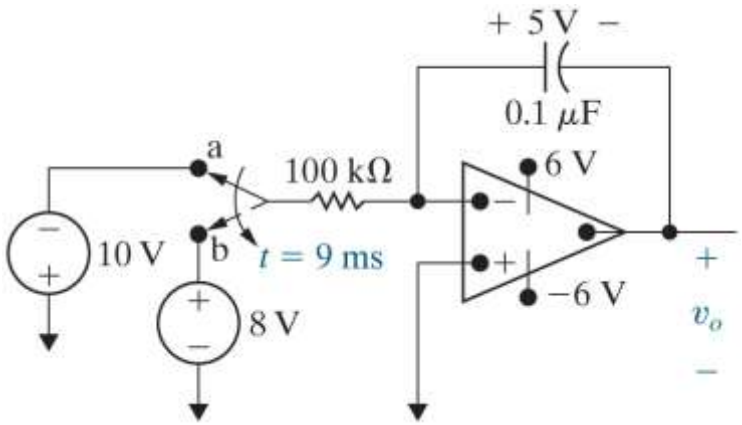
Chapter 7 – Integrating Amplifier (5 of 6)

Example – At the instant the switch makes contact with terminal a in this circuit, the voltage on the capacitor is 5 V. The switch remains at terminal a for 9 ms, then moves instantaneously to terminal b. How many millisecond after making contact with terminal b does the op amp saturate?



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Chapter 7 – Integrating Amplifier (6 of 6)



While the switch is a terminal a, the output voltage is

$$v_o = -\frac{1}{(100\text{ k})(0.1\ \mu)}\int_0^t -10\ dy + (-5)$$
$$= (1000t - 5)\text{ V}.$$

At 9 ms, the output voltage is

$$v_o(9\text{ ms}) = \left[1000(0.009) - 5\right] = 4\text{ V (no saturation)}.$$

After 9 ms, the output voltage is

$$v_o = -\frac{1}{(100\text{ k})(0.1\ \mu)}\int_{9\text{ m}}^t 8\ dy + (4) = -800(t - 0.009) = (11.2 - 800t)\text{ V}.$$

After 9 ms, the output voltage is decreasing from 4 V so saturates at −6 V:

$$11.2 - 800t = -6 \quad \Rightarrow \quad t = 21.5\text{ ms}$$

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